MISHMASH OPENING CONFERENCE · MACHINES SESSION

Building a Norwegian Ecosystem for AI and Creativity

Kilden, Kristiansand · 14 September 2026

THREE QUESTIONS · FOR EVERYONE IN NORWAY

1. Which tools *can* we use?
2. Which tools *should* we (not) use?
3. Which tools do we need to *develop*?

WHO THE MAP IS FOR

- Universities and university colleges
- Research institutes
- Students, bachelor to PhD
- Teachers and schools
- Freelance artists
- Public sector: museums, libraries, broadcasters, municipalities
- Private sector: studios, film, games, startups

CREATIVE AI IS NOT "TRADITIONAL" AI

AI for a task

- one task, one right answer
- batch: throughput matters, latency does not
- large, clean, labelled data
- measured by accuracy on a benchmark
- runs in a data centre, out of sight

AI for creative work

- open-ended: many good answers, or none
- real-time and interactive: latency is the constraint
- small, personal, messy data, often your own
- measured by taste, context and who is asking
- on stage, in the room, on the body: control matters

MAP OF THE INTRODUCTION

STRIPS									
	Rent or run two roads · the middle road	**Own hardware** the ladder · memory	**Shared compute** server · VDI · Fox · Sigma2 · LUMI	**Frontends** chat · code · creative · programmatic	**Models** what runs where · licences	**Who needs what** seven work packages	**Use or develop** off the shelf · new data · new methods	**Holes in the map** who gets access	**The panel** four angles

Rent or run

Rent a service

ChatGPT · Claude · Gemini · Suno · Runway · Midjourney

+ the best models, today
+ zero setup, pay per use
+ works on any device

– no weights, no reproducibility
– your data leave the room
– terms and prices change
– same for everyone, no yours

control: theirs

The middle road

open models, rented GPUs (HF, Mistral, EU providers)

Run it yourself

open models on hardware you or your institution control

+ full control, reproducible
+ private: nothing leaves
+ can be trained on your material

– one step behind the frontier
– you carry the setup
– you pay the energy bill
– needs hardware and time

control: yours

Six questions to ask of any tool

Which model?

Name and version. "AI" is not an answer.

Where does it run?

Your machine, your institution, an EU data centre, the US.

Where do my data go?

Prompt, files, recordings. Stored? Trained on?

Who owns the output?

And can you sell a work made with it?

Same result next year?

Reproducibility for research; continuity for a repertoire piece.

What does it cost?

Per month, per hour of GPU, per kilowatt-hour.

Your own hardware


TinyML · RAVE (real-time audio)
1–8B LLM at 4-bit (≈1–5 GB) · Whisper · small diffusion
14–30B (≈10–20 GB) · SDXL, FLUX · fine-tuning
70B+ (≈40 GB)
memory (RAM, VRAM or unified)
256 KB
Micro-controller
Arduino, ESP32
keyword spotting, gesture classes
100–500 kr
4–16 GB
Single-board
Raspberry Pi, Jetson
small vision/audio, Whisper tiny
1–5 000 kr
6–12 GB
Phone
on-device models
1–3B LLM, camera, mic
5–15 000 kr
8–16 GB
Tablet
any brand
like a phone, bigger screen
5–20 000 kr
16–64 GB
Laptop
any recent model
7–14B chat, Whisper, RAVE on stage
15–40 000 kr
8–24 GB VRAM
Laptop with GPU
RTX 4070–5090 mobile
diffusion, fine-tune small models
25–60 000 kr
32–512 GB
Desktop / workstation
big GPU or big memory
70B local, train small models, serve
40–150 000 kr

parameters × bytes per parameter ≈ memory

7–8B	4-bit	≈ 5 GB	any recent laptop or phone
30B	4-bit	≈ 20 GB	laptop with 32 GB unified, or a 24 GB GPU
70B	4-bit	≈ 40 GB	large unified memory, or two big GPUs
70B	16-bit	≈ 140 GB	a server, or Fox

What each rung is good for

Microcontroller

- keyword spotting, gesture classes
- sensor fusion, milliseconds
- no LLM, ever
- wearables, instruments

Pi / Jetson

- small vision and audio models
- Whisper tiny, YOLO
- an embodied agent in a box
- installations that run for a year

Phone / tablet

- 1–3B on-device LLMs
- camera, mic, sensors
- closed platforms, app stores
- the device every pupil has

Laptop

- 7–14B chat, offline
- Whisper transcription
- RAVE and nn~ on stage
- the studio in a bag

Laptop with GPU

- Stable Diffusion, FLUX
- fine-tune small models (LoRA)
- real-time everything
- hot, loud, one hour of battery

Workstation

- 70B local, video generation
- train RAVE overnight
- serve a whole studio or class
- needs a room and a power bill

School tablet or laptop

- managed device, no GPU
- Feide login, pupils' data
- everything must be hosted
- the most constrained rung

Rule of thumb

- memory decides what fits
- compute decides how fast
- prototype on the smallest rung that works

Shared compute

more compute · more people can share it · more data next to it

Shared server
a GPU box in the lab,
ssh, first come first served

colleagues

VDI
virtual desktop, licensed
software, rarely a real GPU

employees

Institutional HPC
UiO Fox · NTNU IDUN · UiA DGX
UiB and UiT local clusters

own staff, students via a project

National
Sigma2/NRIS: Olivia 448 GH200,
Saga · KI-fabrikken

researchers, by application

European
LUMI (Kajaani), LUMI-AI,
EuroHPC AI Factories

big projects, by call

more waiting · more paperwork · less interactive · batch, not a rehearsal

How you actually get on

Institutional clusters

- UiO Fox: Educloud project, PI adds collaborators
- NTNU IDUN: supervisor requests access
- UiA: DGX servers at CAIR
- UiB and UiT: local HPC via the IT division

NRIS / Olivia

- project application, CPU and GPU hours
- two rounds a year
- any Norwegian research institution
- data storage next to compute

KI-fabrikken

- Sigma2, part of the LUMI AI Factory network
- research, business, public sector
- expertise and support, not only GPUs
- artists and cultural institutions: unclear

LUMI / EuroHPC

- through NRIS or EuroHPC calls
- AMD GPUs: check your software
- large allocations, long lead times

NAIC (paused)

- the low-threshold national GPU service
- funding ended 31 March 2026
- "no-cost hibernation" since April
- successor being procured by Sigma2

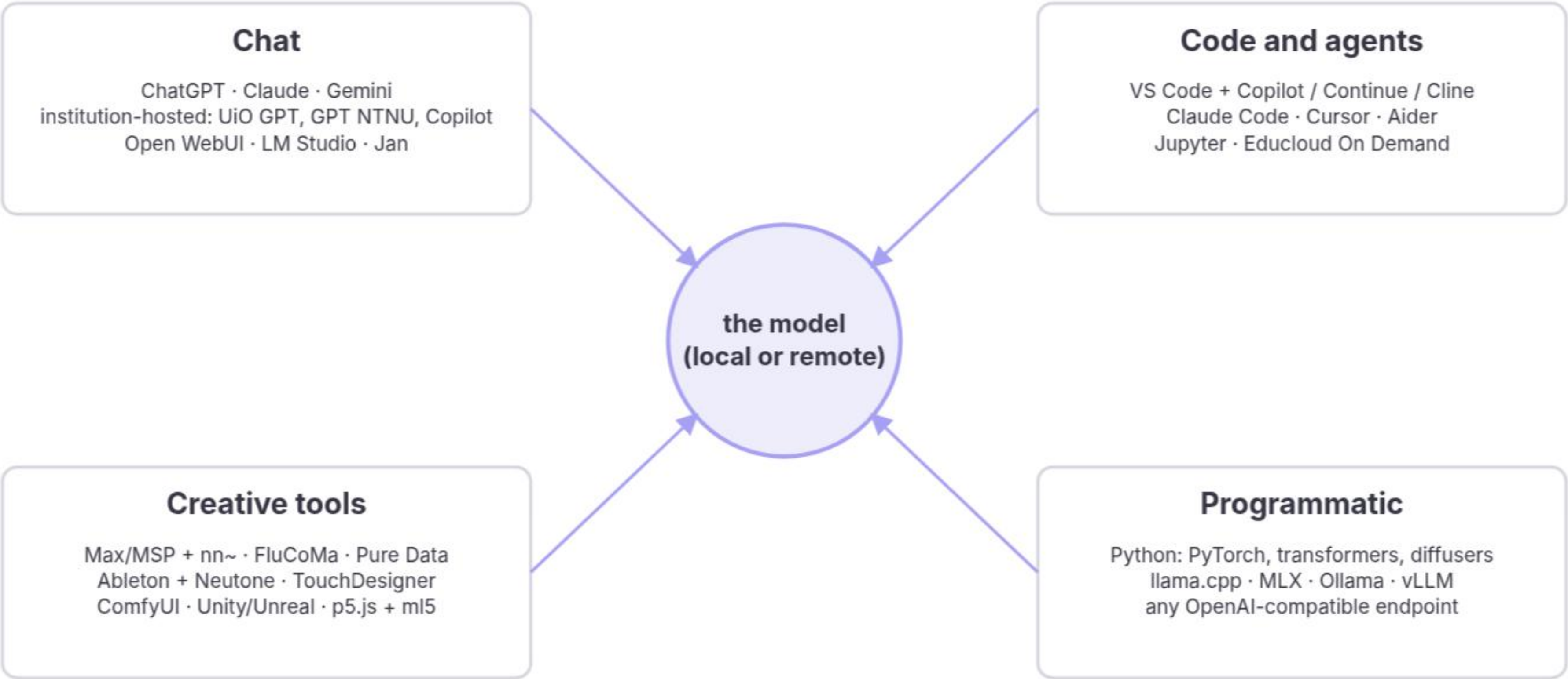
Sensitive data

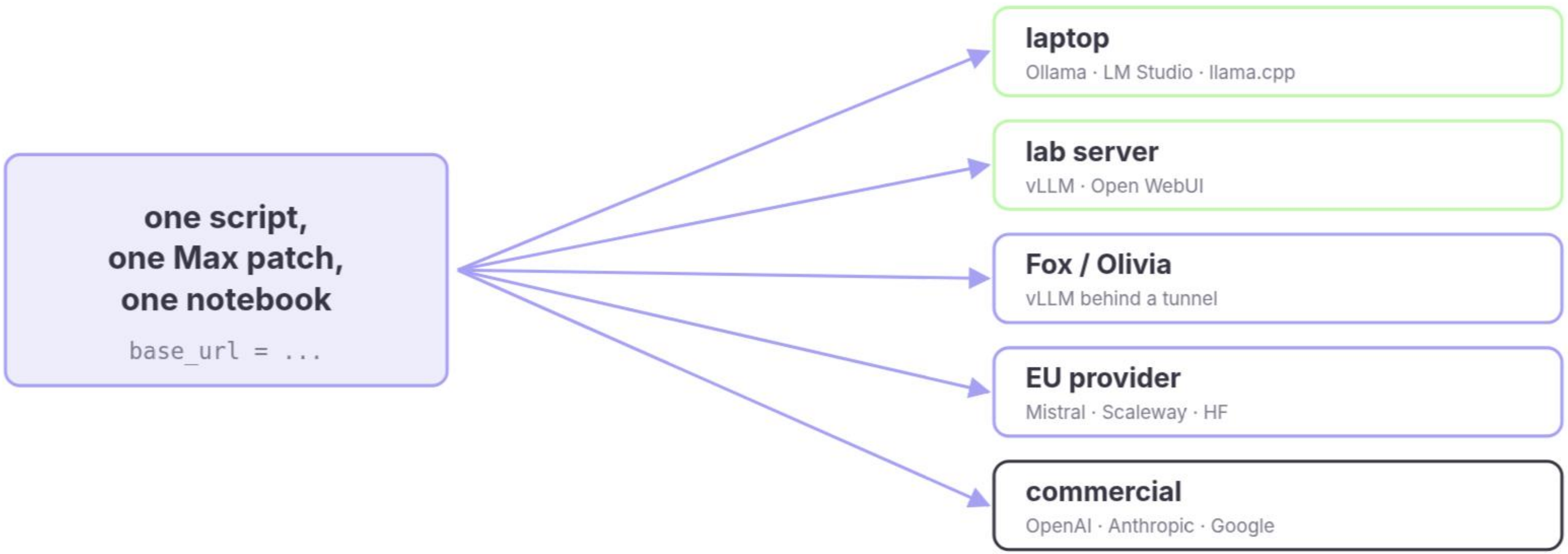
- TSD at UiO, SAFE at UiB, HUNT Cloud at NTNU
- GPU capacity inside is limited
- health and pupils' data live here or offline

Institutions side by side · to be filled in

Institution	Own GPU cluster	Hosted chat / LLM service	Sensitive-data environment	Students get GPUs?	Freelancers / partners get in?
UiO	Fox	UiO GPT	TSD	via project	via a PI
UiB	local HPC (UiB IT)	Copilot	SAFE	?	?
NTNU	IDUN	GPT NTNU	HUNT Cloud	via supervisor	support agreement needed
UiT	local HPC group; hosts NRIS systems	?	?	?	?
UiA	DGX-2 and DGX H100 (CAIR)	?	?	?	?
INN · HiØ · HVL · OsloMet · Nord · Kristiania …	?	?	?	?	?
NMH · KHiO · AHO	none known	?	none	no	?
National Library	DH-lab	?	?	n/a	?
Simula · SINTEF · IFE	eX3 and own	?	?	n/a	?

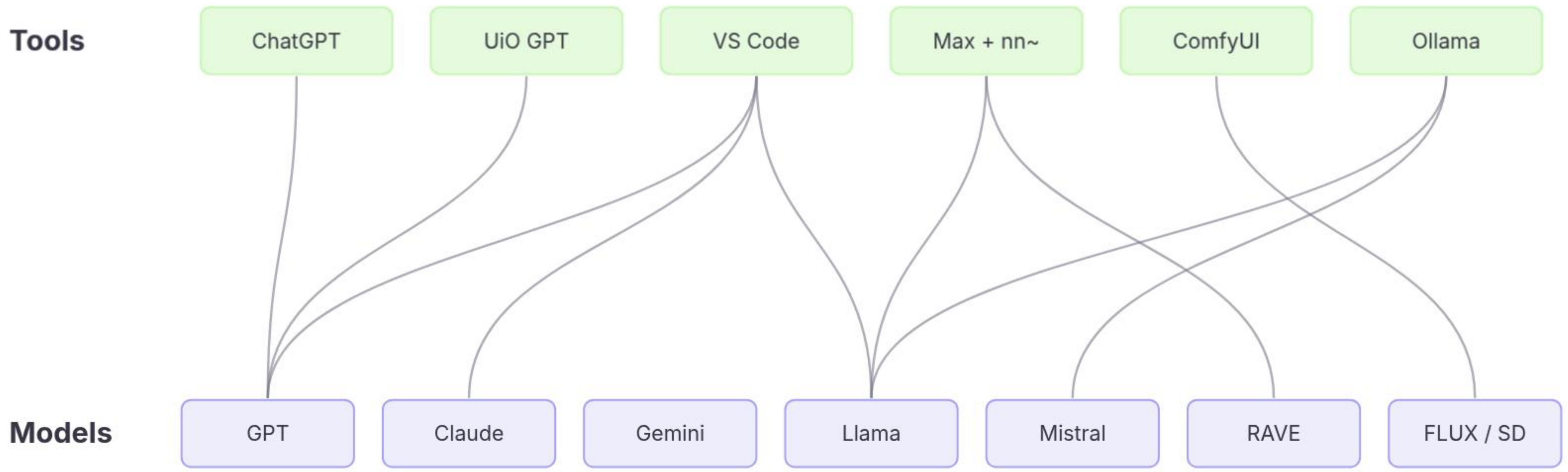
Frontends


Chat
ChatGPT · Claude · Gemini
institution-hosted: UiO GPT, GPT NTNU, Copilot
Open WebUI · LM Studio · Jan
Code and agents
VS Code + Copilot / Continue / Cline
Claude Code · Cursor · Aider
Jupyter · Educloud On Demand
the model
(local or remote)
Creative tools
Max/MSP + nn~ · FluCoMa · Pure Data
Ableton + Neutone · TouchDesigner
ComfyUI · Unity/Unreal · p5.js + ml5
Programmatic
Python: PyTorch, transformers, diffusers
llama.cpp · MLX · Ollama · vLLM
any OpenAI-compatible endpoint



the same OpenAI-compatible API on all of them: change one line, keep the workflow

Tool is not model



Models

Language

Llama · Mistral · Gemma · Qwen · DeepSeek · Phi

Norwegian: NorMistral (UiO) · NorLLM (NorwAI/NTNU) · NB-GPT

Speech

Whisper · faster-whisper · Piper · XTTS · Kokoro

Norwegian: NB-Whisper (National Library)

Music and audio

RAVE (IRCAM) · MusicGen / AudioCraft · Stable Audio Open · ACE-Step · YuE

Norwegian: none

Image

Stable Diffusion · SDXL · FLUX

Norwegian: none

Video

Wan · HunyuanVideo · LTX

Norwegian: none

Vision, motion, multimodal

CLIP · SAM · LLaVA · Qwen-VL · MDM · MotionGPT

Norwegian: none

Licences in three colours

Anything goes

Apache 2.0 · MIT

- Mistral, Mixtral · Qwen3 · DeepSeek · Phi
- NorMistral (UiO) · NB-Whisper
- Whisper · RAVE · CLIP · SAM
- FLUX.1 schnell · Wan 2.1 · YuE · ACE-Step
- use it, sell the work, ship the product

Conditions

Community licences and terms of use

- Llama: "Built with Llama", 700M-user cap, derivatives named Llama
- Gemma: prohibited-use policy passed on
- Stability community licence (SD3.5, Stable Audio Open): free under $1M revenue
- CreativeML OpenRAIL-M (SD 1.5, SDXL): use restrictions travel with the weights
- NorwAI NorLLM: Nordic organisations

Non-commercial

Research only, CC-BY-NC

- **FLUX.1 dev: non-commercial licence for the model**
- **MusicGen: code MIT, weights CC-BY-NC 4.0**
- **many research checkpoints on Hugging Face**
- **fine for a paper; a commissioned work is commercial**

Licence compatibility: what you can do with what

Base model licence	Research paper	Commissioned work, sold	Product or service	Release your fine-tune	Mix with a permissive model	Train a new model on its outputs
Apache 2.0 / MIT Mistral, Qwen3, NorMistral, Whisper, RAVE, FLUX schnell	●	●	●	● any licence, keep the notice	●	●
Llama community Llama 3, Llama 4	●	● "Built with Llama"	◐ under 700M users	◐ same licence, name starts with Llama	◐ the mix inherits Llama terms	◐ allowed, new model named Llama
Gemma terms of use Gemma 2, 3	●	●	◐ prohibited-use policy	◐ terms passed on	◐ the mix inherits the terms	◐ check the terms
Stability community SD3.5, Stable Audio Open	●	◐ free under $1M revenue	◐ same cap, else enterprise licence	◐ same licence	◐ the mix inherits the cap	◐ check the terms
OpenRAIL-M SD 1.5, SDXL	●	●	◐ use restrictions apply	◐ restrictions passed on	◐ restrictions travel with it	◐
Non-commercial FLUX.1 dev, MusicGen weights, CC-BY-NC	●	◐ read it: outputs sometimes yes, the model no	○	◐ non-commercial only	○ the whole release becomes non-commercial	○ usually forbidden

Where to find them, how to trust them

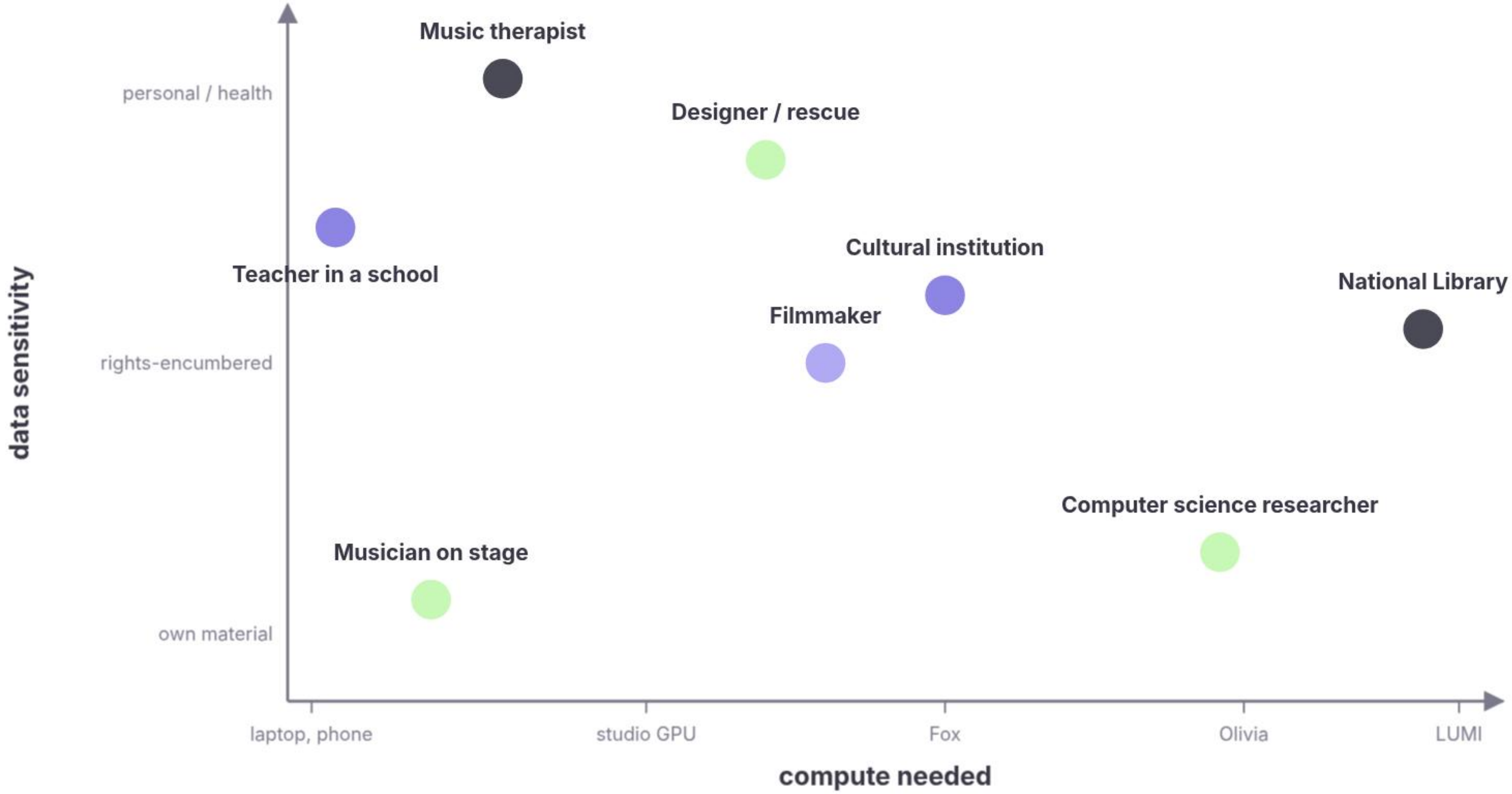
Find

- Hugging Face: weights, model cards, datasets
- Ollama and LM Studio libraries: one-click local
- IRCAM Forum: RAVE and nn~
- the paper's GitHub, for the newest

Trust

- model card: training data disclosed?
- evaluation: on what, by whom?
- reproducible: pinned version, checksum
- energy: training and inference cost stated?

Who needs what


Music therapist
personal / health
Designer / rescue
Teacher in a school
Cultural institution
National Library
data sensitivity
Filmmaker
rights-encumbered
Computer science researcher
Musician on stage
own material
laptop, phone
studio GPU
Fox
Olivia
LUMI
compute needed

WP1 · A musician on stage

prototype small and local

Record

own material,
a few hours of audio,
on the laptop

→

scale up only for training

Train RAVE

overnight on a studio GPU
or a Fox job;
the model is ~30 MB

→

bring it back to the room

Play

nn~ in Max on the laptop,
milliseconds of latency,
nothing leaves the studio

WP2 · A filmmaker

prototype small and local

Sketch

storyboards with an open image model (FLUX, SDXL) on a laptop with GPU

→

scale up only for training

Generate

open video models (Wan, LTX) on a workstation or an EU-hosted provider

→

bring it back to the room

Cut

into the edit; provenance and licence of every model documented for rights

WP4 · A teacher in a school

prototype small and local

Managed devices

tablets or laptops, Feide, no GPU, pupils' data; institution-hosted chat

→

scale up only for training

Offline where it matters

Whisper on the teacher's laptop for transcripts; nothing uploaded

→

bring it back to the room

In the classroom

free, Norwegian, GDPR-safe creative tools; the scarcest resource in the whole map

WP6 · The National Library

prototype small and local

Prepare

petabytes of text, audio, images, film; copyright and public-sector duty

→

scale up only for training

Train

multimodal Norwegian model on Olivia or LUMI, months of allocation

→

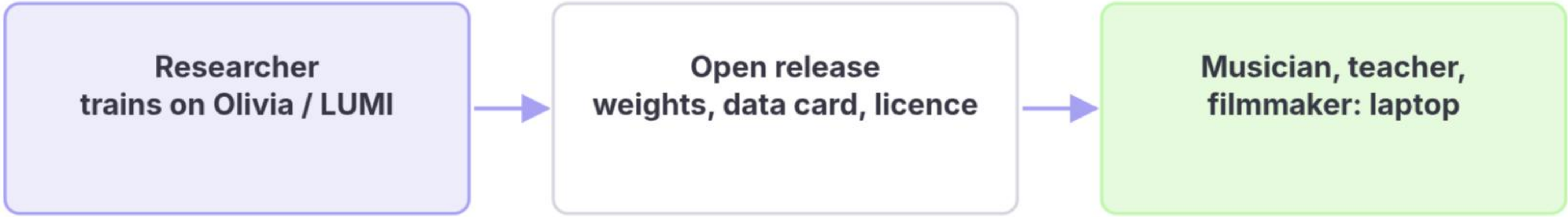
bring it back to the room

Release

open weights on Hugging Face; the musician in WP1 downloads them

Work Package Needs

WP	Typical machine task	Data	Compute	Runs today on	The hole
1 Performances	real-time co-creative agents, RAVE, gesture models	own recordings, sensors	ms; laptop, board, one GPU to train	laptop, studio GPU, Fox	open real-time models; training time without paperwork
2 Processes	image, video, text, audio generation	rights-encumbered	hours on a workstation	commercial tools, workstation	rights-clean open models; EU hosting
3 Health	small models on sessions and signals	health, personal	small, inside a secure zone	TSD, offline laptops	GPUs inside secure zones; consent-ready pipelines
4 Education	AI literacy, classroom tools	pupils' data	none locally; hosted	tablets and laptops, Feide, hosted chat	free, Norwegian, GDPR-safe creative tools
5 Industries	production generation, attribution, audits	contracts, identity	hosted, reliable, auditable	commercial APIs	cultural institutions in KI-fabrikken; open-provider contracts
6 Heritage	transcribe, classify, link, train on the collection	petabytes; copyright, public duty	national or European HPC	NB DH-lab, Olivia, LUMI	multimodal Norwegian models; sustained allocations; storage next to compute
7 Problem-solving	constrained generation, embodied systems	industrial, safety-critical	edge to workstation; simulation	partner systems, Fox	constraint-aware models; access for industry partners



when the loop works: compute upstream, open models downstream

Use or develop

Develop it yourself

two different jobs

Use off the shelf

finished model → any device

hardware: anything that runs inference
who: everyone on the map

time minutes

the catch: their data, their terms

Train an existing architecture on new data

your data → fixed architecture → your model

RAVE on your recordings, a LoRA on your film
hardware: one GPU, sometimes a cluster
who: PhDs, labs, a studio with a GPU

time hours to days, known

the catch: rights to the data

Develop new methods

× × × × ✓

new architectures, objectives, control
hardware: many runs, most fail
who: ML groups on national compute

time ?

the catch: allocations want a number first

Tasks against platforms: a rough guide

Task	Commercial API	Laptop	Laptop + GPU	Workstation	Fox-type cluster	Olivia / LUMI
Chat, writing, code	●	● 7–14B	●	● 70B	◐ overkill	◐ overkill
Transcription (Whisper)	●	●	●	●	◐ batch	◐ overkill
Image generation	●	◐ slow	●	●	◐ batch	◐ overkill
Video generation	●	○	◐ short clips	●	◐ batch	◐
Real-time audio on stage	○ latency	●	●	●	○	○
Train a small model (RAVE, classifier)	○	◐ slow	●	●	●	◐ overkill
Fine-tune an LLM (7–14B)	◐ data leave	○	◐ LoRA	●	●	●
Train from scratch (NB-scale multimodal)	○	○	○	○	◐ small	●
Sensitive data (health, pupils)	○	● offline	● offline	● offline	◐ TSD-type only	○

By person: a plausible stack today

Who	Everyday	Creative work	Training	The catch
Musician on stage	laptop, local chat	Max + nn~, RAVE	studio GPU overnight, or a friend with Fox	no institution: no Fox, no NRIS
Filmmaker	commercial tools	ComfyUI, open video models on a workstation	LoRA on own footage	rights of the base model
Teacher in a school	Feide-gated hosted chat	browser tools only	none	no Norwegian, GDPR-safe creative tool exists
Music therapist	offline laptop	small local models	inside TSD, if GPUs	consent and secure-zone capacity
Bachelor student	Ollama + Open WebUI	Whisper, notebooks	Colab-type, or a course allocation	depends entirely on the institution
PhD / ML researcher	laptop	prototypes locally	Fox, then NRIS, then LUMI	allocation rounds and the paperwork
Cultural institution	commercial APIs	hosted open models with a contract	KI-fabrikken, if they qualify	nobody has defined them as a user category
Startup / studio	commercial APIs	own GPUs or cloud	KI-fabrikken as "business"	cost, and no research partner
National Library	own lab	own models	Olivia, LUMI	sustained multi-year allocation

Holes in the map

Who	Own hardware	Institutional servers, VDI, Fox-type	National HPC NRIS, Olivia	KI-fabrikken	Hosted open models EU providers	Commercial APIs
Researchers, big universities UiO · UiB · NTNU · UiT	yes	yes, varies	by application	yes	own budget	own budget
Researchers, other universities and colleges UiA · INN · HiØ · HVL · NMH · KHiO · AHO · Kristiania ...	yes	often little or none	by application	yes	own budget	own budget
Research institutes	yes	own labs	by application	yes	own budget	own budget
Students, bachelor to PhD	own machine	via a supervisor	via a project	?	rarely	own budget
Teachers and schools	tablets, laptops, no GPU	no	no	no	no	Feide-gated, limited
Freelance artists	if affordable	no	no	?	own budget	own budget
Public sector museums · libraries · NRK · municipalities	yes	own IT, no GPUs	no	"public sector"	budget, procurement	budget, procurement
Private sector studios · film · games · startups	yes	own	only with a research partner	"business"	budget	yes

Observations to argue with

- The academic infrastructure is built for batch science, not for a rehearsal room.
- The people who most need cheap, controllable, rights-clean models have the least access.
- Access depends on which institution you sit in, not on what you need.
- Open Norwegian models exist for language and speech, and for nothing creative.
- The whole Norwegian cultural heritage sits in one collection. Nobody has trained a multimodal model on it.
- Nobody has defined "artist", "teacher" or "cultural institution" as a user of national compute.
- The low-threshold national service (NAIC) is paused; its successor is not yet here.

Over to the panel

1. Which tools *can* we use?
2. Which tools *should* we (not) use?
3. Which tools do we need to *develop*?

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